How Speakers Turn Waves Into Sound

01 How can a wave become a sound?



A speaker gives a smooth signal a moving cone.

Look closer. Find the smooth wiggle beside the speaker.

A little explanation. Sound begins with vibration, a back-and-forth motion. A speaker turns a changing electric signal into motion. A pure note can follow a sine wave's smooth wiggle. Music uses many wiggles together.

Try this. Draw a smooth wave, then move your hand gently back and forth to follow its rhythm.

02 What makes the cone wiggle?



The coil moves the cone, and the cone moves air.

Look closer. Look for the wide paper cone on the speaker.

A little explanation. A speaker has a cone made of paper or plastic. A wire coil is attached behind it, near a magnet. The changing electric signal makes the coil push and pull against the magnet. The cone moves with it.

Try this. Use the diorama to follow the coil, the cone, and the air in that order.

03 Can you see a vibration?



A vibrating surface can make rice hop.

Look closer. Look just above the tight wrap.

A little explanation. Rice can hop when the surface under it vibrates. Try a toy drum, or a bowl with plastic wrap pulled tight across its top. A tiny tap makes the motion easier to spot.

Try this. Place a pinch of rice on the tight wrap or a toy drum. Hum or sing softly nearby and look closely. If the rice stays still, gently tap the bowl's side or the drum rim. Draw what changes. Keep Big beside you, sounds gentle, and rice on the table, away from mouths and ears. Leave devices closed; use no wall power or soldering, and let Big hold any magnet.

04 How does the wiggle reach your ear?



Pressure waves make an eardrum vibrate.

Look closer. Follow the air-wave marks toward the ear.

A little explanation. The cone squeezes air, then lets it spread out. These pressure changes travel as a wave. At your ear, the eardrum vibrates. Faster wiggles make a higher pitch. Bigger wiggles make a louder sound.

Try this. Draw a path from speaker to air to ear. Add one question about where the wiggle goes next.